

Forward Model

INITIAL CONDITIONS

GRAVITY

FINAL FIELDS

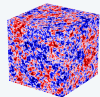
COMPRESSION

INFERENCE

From prior: θ

Linear $P_L(k)$

DiscoDJ



δ_{ic}
initial conditions

2LPT + PM

scatter

uniform weights = 1

Lagrangian bias $w(\mathbf{q})$

$$1 + b_1 \delta + b_2 (\delta^2 - \langle \delta^2 \rangle) + b_{s^2} (s^2 - \langle s^2 \rangle) + b_{n^2} \nabla^2 \delta$$

δ_{dm}
dark matter



δ_g
biased tracer



Field compression

CNN / ViT

BFast

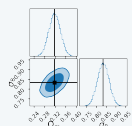
$$P_{0,2,4}(k) + B_0(k_1, k_2, k_3)$$

Fisher
forecast

Summary compression

none / MLP / PCA

SBI



posterior $p(\theta | \mathbf{d})$